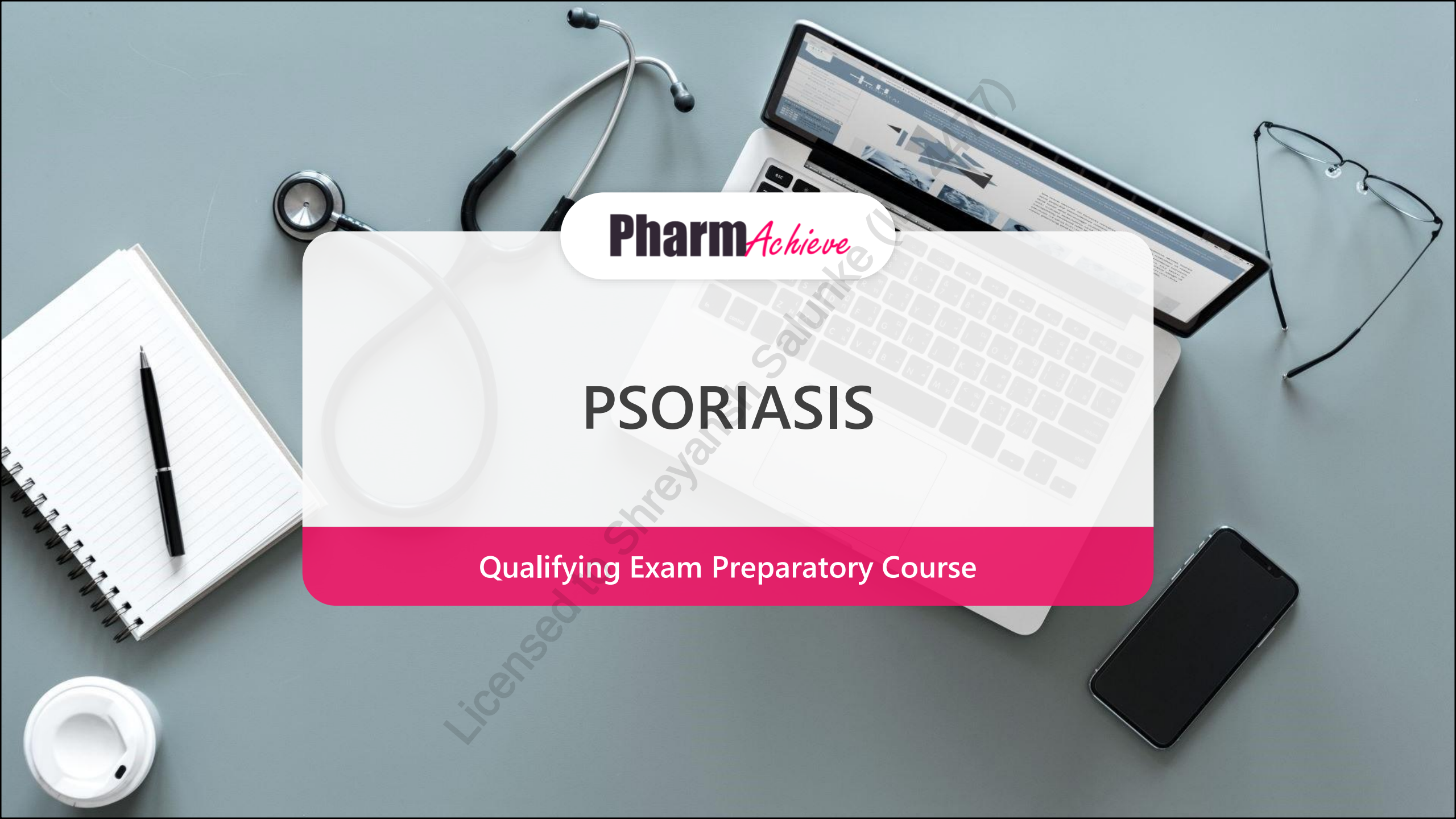


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

PSORIASIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE Adverse Effects

AM Ante Meridiem (Morning)

BID Twice A Day

BRM Biologic Response Modifiers

BSA Body Surface Area

CYP 450 Cytochrome P450

DLQI Dermatology Life Quality Index

GI Gastrointestinal

IgG1 Immunoglobulin G1

IL Interleukin

N/V/D Nausea/Vomiting/Diarrhea

NB-UVB Narrow-band UVB

NPF National Psoriasis Foundation

NSAIDs Nonsteroidal Anti-Inflammatory Drugs

PASI Psoriasis Area and Severity Index

PDE-4 Phosphodiesterase 4

PGA Physician's Global Assessment

PUVA Psoralen Plus Ultraviolet A Therapy

QHS Quaque Hora Somni (Every Night At Bedtime)

TB Tuberculosis

Th1 T Helper Type 1 Cell

Th17 T Helper Type 17 Cell

TID Three Times A Day

TNF Tumor Necrosis Factor

URTI Upper Respiratory Tract Infection

UVA Ultraviolet A

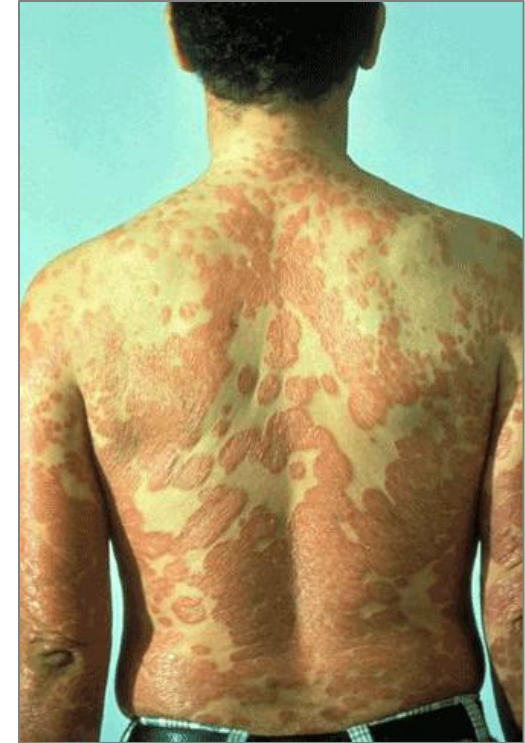
UVB Ultraviolet B

OBJECTIVES

- 1 Define and classify psoriasis
- 2 Discuss the pathophysiology, risk factors, and clinical presentation of psoriasis
- 3 Determine when self-treatment is appropriate
- 4 Discuss the treatment and management of psoriasis

DEFINITIONS AND PATHOPHYSIOLOGY

- Psoriasis is a **systemic chronic disease with physical and psychological manifestations**
- It is an **inflammatory autoimmune disease involving skin, joints, and other areas**
- Psoriasis involves **the innate and adaptive immunity**:
 - Innate immunity: dendritic cells, macrophages, and neutrophils which produce cytokines
 - Adaptive immunity: activated T-cells which includes T-helper type 17 cells (Th17) and T-helper type 1 cells (Th1)
 - **T-cell driven inflammation (plays a major role): keratinocyte overgrowth, angiogenesis, inflammation**



RISK FACTORS AND TRIGGERS

Risk factors

- Genetics
- Ethnicity
- Smoking
- Obesity
- Vitamin D deficiency
- Alcohol consumption

Triggers

- Infections (post-streptococcal flares of guttate psoriasis)
- Drugs (e.g. beta-blockers, lithium, antimalarials, oral steroids, NSAIDs)
- Stress
- Injury/trauma
- Low humidity
- Season (most patients worsen and experience flare-ups during the **cold** season)
- Post-partum period

CLINICAL PRESENTATION - PLAQUE PSORIASIS

General presentation	• Red and erythematous plaques, silver scales/flakes, pain, pruritus, diameter <1 cm to 10 cm, occurs on scalp, trunk, and extremities
Nail psoriasis	• Nail bed and nail matrix: discolouration, pitting, thickening, splintering, nail plate removed from nail bed
Sebopsoriasis	• Face: thin, red, sharp bordered lesions, appears greasy
Scalp psoriasis	• Plaques with thick white-silver scaling • May be found in area above the ears, scalp margin
Flexural/inverse psoriasis	• Skin folds: thinner, shiny plaques • May involve genitals, groin, natal cleft, under the breasts, axillary
Palmoplantar psoriasis	• Palms of hands or soles of feet (appearance varies): redness and scaling with no definable plaques, large plaques, poor scaling

CLINICAL PRESENTATION - NON-PLAQUE PSORIASIS

Guttate psoriasis

- Abrupt eruption of small papules and plaques, drop-like appearance, commonly occurs after respiratory infection (mainly streptococcal infection)

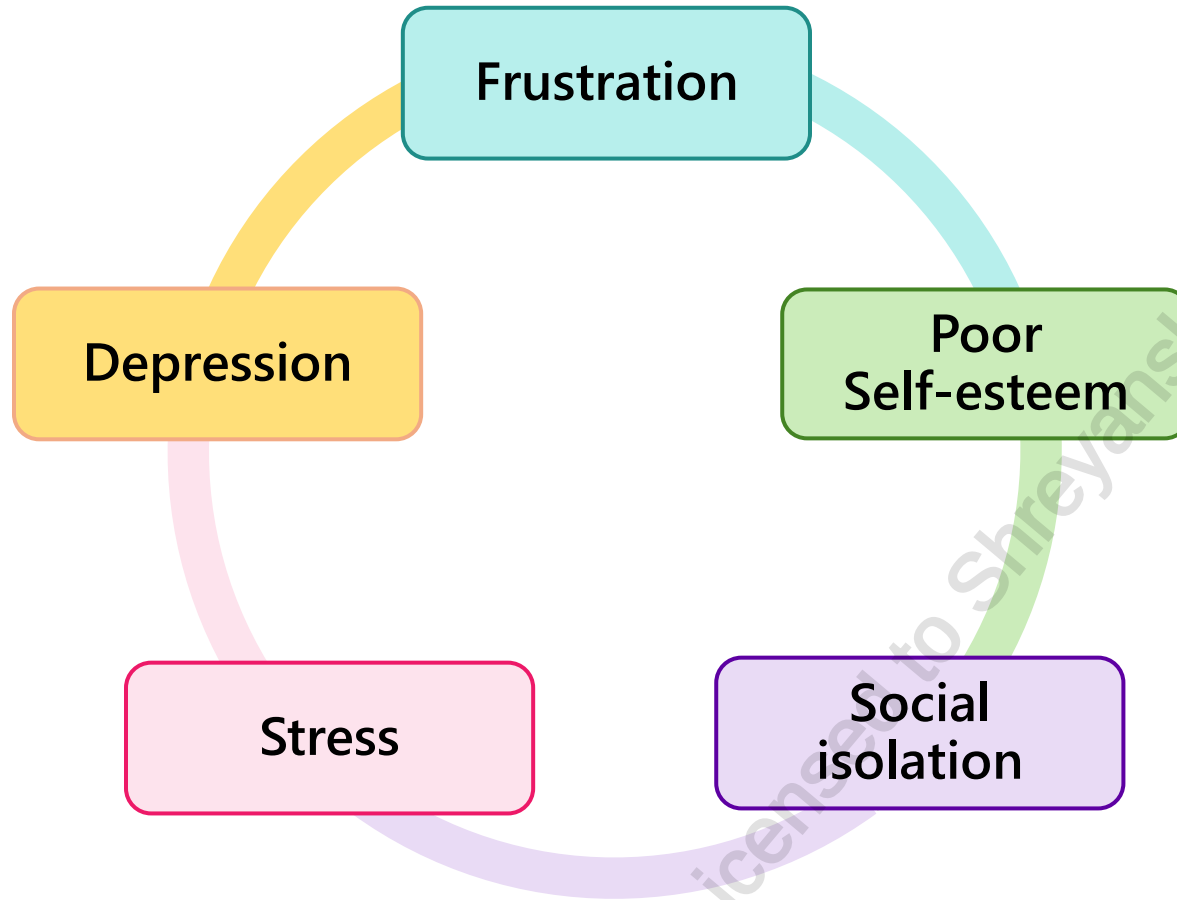
Erythrodermic psoriasis

- Acute or chronic inflammatory patches, edema, desquamation (peeling)
- Covers a large portion of the body (frequently covers about 90% BSA)
- Increase risk of complications (e.g. sepsis, electrolyte imbalance)

Pustular psoriasis

- May have systemic symptoms including fever, malaise, leukocytosis, diarrhea, hypocalcemia
- Eruption of pustules around inflamed skin or plaques
- Palmoplantar pustulosis: occurs specifically on palms of hands and soles of feet
- May occur in isolation or secondary to plaque psoriasis
- Potentially severe complications

CLINICAL PRESENTATION - PSYCHOLOGY



Patient surveys by the National Psoriasis Foundation (NPF) (2001 and 2008) show:

- Disease was a significant problem in everyday life
- The negative effect on physical, psychological and social dimensions of life can be greater than those from life-threatening illnesses, such as myocardial infarction

SUMMARY CHART → PLAQUE PSORIASIS

	Psoriasis subtype	Affected area	Characteristics
PLAQUE	Flexural/inverse	Skin folds: e.g. groin, axillary, genitals	Involved in 90% psoriasis cases: • Red, erythematous plaques • Silver scales • Sharp borders
	Nail	Nail bed and/or nail matrix	
	Scalp	Scalp	
	Palmoplantar	Palms of hands, soles of feet	
	Sebopsoriasis	Face	

SUMMARY CHART → NON-PLAQUE PSORIASIS

	Subtype	Affected area	Characteristics
NON-PLAQUE	Guttate Psoriasis	Trunk, limbs, face	Eruption of small papules triggered by respiratory infections, especially streptococcal infections
	Pustular Psoriasis	• Palmoplantar pustulosis: palms of hands, soles of feet • Can include trunk or extremities	Pustules around plaques, inflamed skin
	Erythroderma Psoriasis	Covers large amounts of total body surface area, often >90% BSA	Inflammatory patches, edema, desquamation → High risk of complications

EVALUATION AND ASSESSMENT TO ASSESS SEVERITY

Body Surface Area (BSA)

- Use the palm method (palm including fingers) to represent 1% of BSA
- National Psoriasis Foundation:
 - Mild <3% BSA
 - Moderate 3-10% BSA
 - Severe >10% BSA

Psoriasis Area and Severity Index (PASI)

- Symptoms, coverage size, and location(s) of psoriasis

Physician's Global Assessment (PGA)

- Static PGA: Evaluate clinical presentation at a point in time
- Dynamic PGA: Evaluate response to therapy by observing change in presentation

Dermatology Life Quality Index (DLQI)

- Patient questionnaire to assess views on treatment, lifestyle, and clinical manifestation

DIAGNOSIS

- All psoriasis should be diagnosed by a physician before treatment
- Psoriasis is diagnosed mainly through **physical examination and history**
- Full examination includes **scalp, nails, anogenital skin**
 - Looking for well-demarcated plaques and scales (silver)
 - Koebner's phenomenon (plaques along lines of trauma)
 - Auspitz sign (pinpoint bleeding after removal of scale)
- Insufficient evidence may require a skin biopsy
- Review the patient's family history and medications

GOALS OF THERAPY

- ✓ Provide symptom relief
- ✓ Improve quality of life
- ✓ Prevent exacerbations
- ✓ Avoid treatment-related adverse effects

NON-PHARMACOLOGICAL MEASURES

Avoid triggers	Scalp	• Use non-irritating, fragrance-free shampoo • Wash hair at least 2-3x/week
Moisturizers	Moisturize involved areas and remove scale debris	
Humidifier	Folds	• Wear loose-fitting clothing • Choose moisture-wicking fabrics
Occlusive Dressings • Hydrocolloids: expensive • Hydrogels: high water content	Reduce irritation, maceration, skin trauma	
	Nails	• Keep nails short • Avoid aggressive manicures or pedicures
	Decrease risk of nail separation by trauma	

THERAPEUTIC REGIMENS

Monotherapy (single agent)

- Limited side effects, decreased cost, improved adherence
- May be inadequate therapy
 - Additional or change of therapy may be necessary

Combination therapy (≥ 2 agents)

- More effective than monotherapy
- Lower doses are used: minimizes toxicity, improves efficacy
- May taper down and discontinue one agent once the condition is controlled

Rotation therapy (rotate therapies periodically)

- Intervals range from 1-2 years; limits cumulative toxicity

Sequential therapy

- Initial therapy: rapidly effective agent for quick improvement
- Transition to safer maintenance therapy
- Minimize side effects

PHARMACOLOGIC THERAPY - CHOOSING A TOPICAL VEHICLE

Topical therapy:

- Vehicle may affect adherence and drug potency (ointment versus cream)
- First-line therapy for mild-to-moderate cases
- Adjunctive therapy in moderate to severe psoriasis

Vehicle	Characteristics
Ointment	• Maintains hydration, good penetration • May stain clothes (oily) • Preferred in dry and cold environments
Creams	• Spreads easily, absorbed and dries quickly • Useful for skin folds • Works well in warm environment
Lotions	• Requires more drying than creams • Suitable for thinner-skinned areas, useful for the scalp and skin folds • Works well in warm environment
Foams	• Easy to apply, requires more drying • Useful for the scalp
Gels	• Cooling effect • Drug delivered via evaporation • Useful for the scalp

PHARMACOLOGIC THERAPY - NON-PRESCRIPTION TOPICAL THERAPY

Topical steroids

- Hydrocortisone 0.5%: low potency, useful for very mild cases
- 1st line therapies for mild-moderate plaque psoriasis
- Side effects: thinning skin, tachyphylaxis (tolerance), skin atrophy, contact dermatitis

Petrolatum (adjunctive)

- An emollient option to moisturize the skin (especially after showering/bathing)

Salicylic acid (adjunctive)

- Increases penetration of topical steroids; blocks UVB radiation
- **Caution: salicylate toxicity may occur in patients that require >20% BSA**

PHARMACOLOGIC THERAPY - NON-PRESCRIPTION TOPICAL THERAPY

Lactic acid (adjunctive)

- May be an option for scalp psoriasis

Coal tar shampoo (adjunctive)

- Anti-inflammatory and anti-proliferative effects
- Adjunctive therapy for scalp psoriasis and scalp sebopsoriasis
- 0.5-1% applied daily and left in place for 5-10 minutes before rinsing

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL THERAPY

OVERVIEW

Corticosteroids

Vitamin D Analogues

Retinoids

Calcineurin Inhibitors

Phosphodiesterase-4 Inhibitor

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL THERAPY

CORTICOSTEROIDS

- 1st line therapy for mild-moderate plaque psoriasis
 - Low-potency: ideal for face or folds
 - Moderate-potency: more commonly used on extremities
 - High-potency: more resistant cases
- Side effects: thinning skin, tachyphylaxis (tolerance), skin atrophy, contact dermatitis
- Avoid applying to eye area due to an increased risk of glaucoma

Low potency	Moderate potency	High and ultra-high potency
Hydrocortisone 0.5%, 1%, 2.5%	• Betamethasone valerate 0.05% 0.1% • Hydrocortisone 17-valerate 0.2%	• Betamethasone dipropionate 0.05% • Clobetasol 17-propionate 0.05% (ultra-high) • Fluocinonide 0.05% • Mometasone furoate 0.1% • Betamethasone/salicylic acid (combination with a high-potency keratolytic agent)

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL THERAPY

VITAMIN D ANALOGUES

Calcitriol, calcipotriol, and calcipotriol/betamethasone

- Combination calcipotriol/betamethasone is superior to monotherapy
- Scalp: apply gel to scalp QHS and rinse in AM
- Body: apply ointment or gel to the affected area once daily
- Calcitriol is 1st line for mild plaque psoriasis (monotherapy or combination)
- Well tolerated and no tachyphylaxis
 - Side effects: skin irritation, hypercalcemia (rare)

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL THERAPY

RETINOIDS

Tazarotene

- Monotherapy option now only indicated for acne, but available in combination with halobetasol which is applied once a day
 - 0.01% halobetasol propionate + 0.045% tazarotene
- Not to use on face, scalp, or skin folds
- Increases efficacy of UVB radiation
- Avoid in pregnancy (use effective contraception while using retinoids)
- Side effects: skin irritation, photosensitivity

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL AGENTS

CALCINEURIN INHIBITORS

Tacrolimus, pimecrolimus

- 2nd line therapy that is NOT approved for psoriasis treatment
- Useful in thin-skinned areas e.g. face and folds
- Potential steroid-sparing therapy
- Not associated with cutaneous atrophy
- Uncertain concern over long-term use and risk of cancer (use minimal amount for the shortest duration necessary)

PHARMACOLOGIC THERAPY - PRESCRIPTION TOPICAL AGENTS

PHOSPHODIESTERASE 4 INHIBITORS

Roflumilast

- Zoryve® 0.3% foam is currently approved for the topical treatment of plaque psoriasis, including in the intertriginous areas (skin folds or creases)
- Approved in adults and children ≥ 12 years
- Demonstrated improvements in scalp and body psoriasis
- May also benefit pruritus associated with psoriasis
- Well tolerated
- Side effects: application-site pain, GI disturbance (n/v/d), headache

PHARMACOLOGIC THERAPY - PRESCRIPTION SYSTEMIC THERAPY

OVERVIEW

Immunosuppressive Systemic Agents

Oral Retinoids

Biologic Response Modifiers

PHARMACOLOGIC THERAPY - ORAL RETINOIDS

ACITRETIN

- Enhanced efficacy when combined with other treatments (e.g. phototherapy, calcipotriol)
- Less risk of toxicity compared to immunosuppressive therapy
- Ethanol increases metabolism to major teratogen; women of childbearing age to abstain from alcohol during treatment and for 2 months after cessation of therapy
- **Avoid in pregnancy** (two forms of birth control recommended during and for **3 years** after cessation of therapy – **progesterone (minipill) is not reliable**)
- Increased risk of liver injury when used concurrently with methotrexate
- Caution when combining with tetracyclines due to possible risk of pseudotumor cerebri
- Caution with high doses of vitamin A due to toxicity (fat-soluble vitamin)
- Side effects: dry mucosa, headache, hyperlipidemia, photosensitivity, psychiatric issues, hepatotoxicity, arthralgias/myalgias, alopecia

PHARMACOLOGIC THERAPY - BIOLOGIC RESPONSE MODIFIERS

- Effective for chronic moderate-severe psoriasis
- Expensive (e.g. >\$20,000/year)
- Avoid live vaccinations during biologic therapy
- Increased risk of serious infections with concurrent immunosuppressants
- Side effects: injection-site reactions, upper respiratory tract infections, reactivation of latent TB, risk of malignancy

Adalimumab	Bimekizumab	Brodalumab
Certolizumab pegol	Etanercept	Guselkumab
Infliximab	Ixekizumab	Risankizumab
Secukinumab	Ustekinumab	Tildrakizumab

PHOTOTHERAPY

- PUVA: use psoralen and UVA phototherapy
- Topical or oral psoralen treatment is used prior to UVA
- Combination therapy lowers the dose of therapeutic drug and reduces UV exposure
- Take photosensitizing drugs after phototherapy to prevent burns
- Treatment should be on a regular schedule: 3-5 times/week where fewer treatments per week will require a longer duration overall
- Side effects: photosensitivity, skin aging, increased risk of cancer

PHOTOTHERAPY

- UVB
- Broad band UVB: combined therapy lowers doses of systemic therapy
- Narrow-band UVB (NB-UVB): effective monotherapy or combination therapy
- Excimer laser treatment that emits monochromatic 308-nm radiation that restricts exposure to psoriatic skin
- Fewer adverse effects because of limited treatment area; side effects include photosensitivity, skin aging, increased risk of cancer
- Take photosensitizing drugs after phototherapy to prevent burns
- Treatment should be on a regular schedule (3-5 times per week), where fewer treatments per week will require a longer duration overall

PHARMACOLOGIC THERAPY - INTRALESIONAL CORTICOSTEROID INJECTION

TRIAMCINOLONE

FYI

- Useful in very resistant plaque psoriasis and nail psoriasis
- Side effects: very painful upon injection

PLAQUE PSORIASIS



MILD TO MODERATE PSORIASIS THERAPEUTICS

Topical medications and emollients

- 1st line: topical steroids, calcipotriol, calcitriol, calcipotriol + betamethasone dipropionate
- May be used as monotherapy or combination

Topical agents + phototherapy

- Tar + UVB or NBUVB, PUVA

Topical + systemic agents

MODERATE-SEVERE PSORIASIS THERAPEUTICS

Systemic agents

- +/- topical agent or phototherapy
- May be used as monotherapy or combination

OR

Biologics

- +/- topical agent or phototherapy
- Consider cost and safety

SPECIAL AREAS - PLAQUE PSORIASIS

Facial and flexural psoriasis

- Face: hydrocortisone 1% cream
- Other therapies: ketoconazole (fungal infection of folds), tacrolimus, pimecrolimus

Nail psoriasis

- Attempt topical therapy (trimming nails reduces the risk of traumatic separation from the nail bed and facilitates application of topical treatments)
- Intralesional corticosteroid injection: for severe nail psoriasis, injection may be painful
- Systemic therapy
 - Avoid in routine use of isolated incidents
 - Useful when treating another type of psoriasis
- Phototherapy: limited evidence of efficacy
 - Avoid in routine use of isolated incidents
 - Useful when treating another type of psoriasis

MODERATE-SEVERE PSORIASIS - ORAL THERAPY

Acitretin (PO retinoid)

- 1st line treatment of chronic palmoplantar or pustular psoriasis in non-childbearing patients, particularly in patients with severely sun-damaged skin
- Might be more effective if combined with other treatments
- Adverse effects are teratogenicity, mucocutaneous dryness, and elevated triglyceride levels
- Limitations are slow onset of action and residual plaque psoriasis

Cyclosporine (PO calcineurin inhibitor)

- 1st line treatment for acute, severe pustular psoriasis or erythrodermic psoriasis
- Intermittent use for no more than 12 weeks to control flare-ups is recommended
- Adverse effects include cumulative renal toxic effects, hypertension, hypertriglyceridemia, and higher risk for squamous and basal cell carcinomas

MODERATE-SEVERE PSORIASIS: ORAL THERAPY

Methotrexate (PO inhibitor of folate biosynthesis)

- 1st line treatment for moderate-severe plaque psoriasis
- Less effective when compared with cyclosporine but may be used continuously
- Adverse effects include cumulative liver toxic effects, bone marrow suppression, and teratogenicity
- Pretreatment and follow-up liver biopsies might be recommended in some cases
- Folate supplementation reduces liver toxicity, but not efficacy
- Contraception for men and women should be used during the course of therapy and for a limited time following cessation of therapy

Apremilast (PO PDE-4 inhibitor)

- Modulates the production of several inflammatory mediators at the level of mRNA expression by altering cAMP levels in keratinocytes and T-cells
- Indicated for moderate-to-severe plaque psoriasis in adults
- Dosed at 30 mg PO BID (dosed once daily in renal impairment)
- Side effects include nausea, upper respiratory infection, headache, and diarrhea

MODERATE-SEVERE PSORIASIS - INJECTABLE AGENTS

TNF-alpha inhibitors

- **Safety concerns include serious infections, autoimmune conditions, and malignancies**
- Etanercept and adalimumab are administered subcutaneously whereas infliximab is administered intravenously
- Adalimumab vs methotrexate or placebo had greater improvement and lower rate of adverse events

Ustekinumab (Stelara®)

- Human monoclonal antibody that binds to shared p40 subunit of IL-12 and IL-23
- **May be used first-line in moderate-severe plaque psoriasis**
- Administered subcutaneously
- **Possible risks of infection and neoplasm – not a lot of long-term data available**

Secukinumab (Cosentyx®)

- Human IgG1 monoclonal antibody that selectively binds to and inactivates IL-17A
- Interleukin 17A plays a key role in the pathogenesis of plaque psoriasis
- **Indicated for moderate-to-severe plaque psoriasis and psoriatic arthritis in adults**

SPECIAL AREAS - NON-PLAQUE PSORIASIS

Guttate psoriasis

- **First-line: phototherapy, particularly NB-UVB**
- Adjunct therapy: topical steroids or topical vitamin D analogues
- Treatment failure: attempt systemic therapy
- Antibiotics if streptococcal infection

Pustular psoriasis

- **First-line: Acitretin, cyclosporine, methotrexate, infliximab**
- Alternative first-line: secukinumab, ixekizumab, guselkumab
- Second-line: PUVA, adalimumab, etanercept
- Adjunct: topical therapy

TREATMENT IN SPECIAL POPULATIONS

Pregnancy

- Options are **topical corticosteroids, UVB**, and cyclosporine (**reserve for severe cases**)
- **Calcitriol, anthralin, and PUVA should be avoided if possible due to lack of evidence of safety**
- Biological agents might be needed if benefits > risks
 - certolizumab pegol has minimal placental transfer

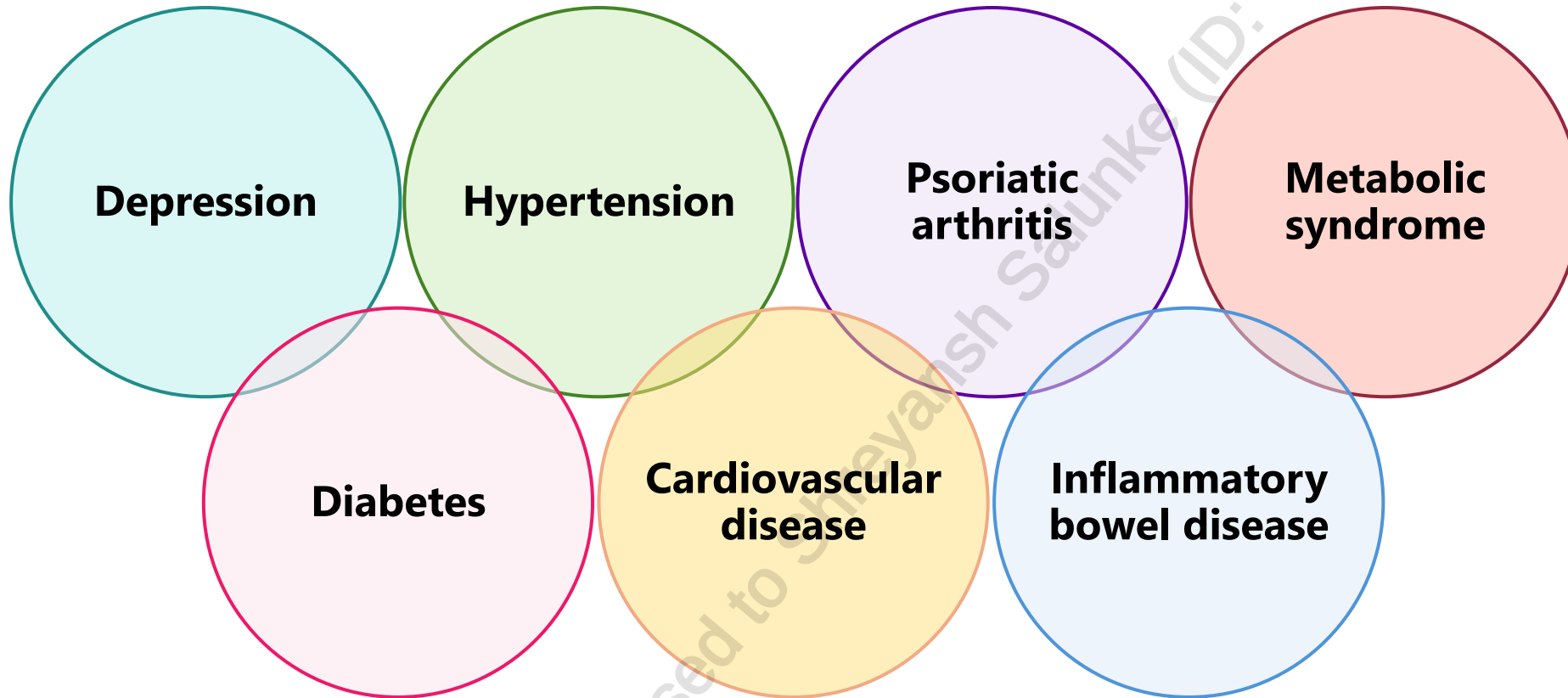
Elective surgery

- Prior to surgery, discontinuation of TNF-alpha inhibitors for 1 week prior and 1 week after should be considered to reduce complications

Inactivated vaccines

- May have reduced efficacy
- Live or live-attenuated vaccines may cause infections in patients receiving systemic immunosuppressive treatment

PSORIASIS IS ASSOCIATED WITH A HIGHER INCIDENCE OF COMORBIDITIES



MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Disease control: e.g. erythematous plaques and/or white-silver scales, pruritus, nail discoloration/pitting	Significant improvement	1-3 months*

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
AEs of topical therapy (skin atrophy, contact dermatitis)	Prevent or minimize	Duration of therapy
AEs of systemic therapy (N/V/D, headaches, injection site reactions)		
Serious AEs of systemic therapy (e.g. allergic reactions, URTIs, malignancies)	Prevent	

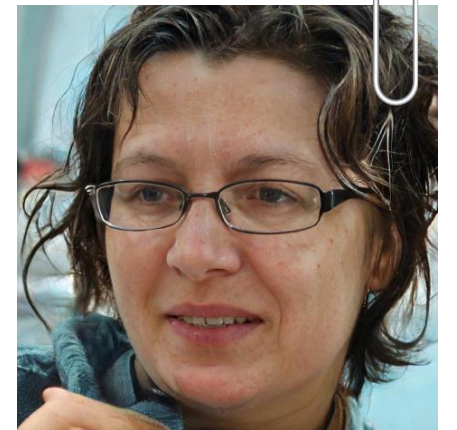
TAKEAWAY

- Psoriasis has no cure and managing the disease is the primary goal
- Be empathetic and understanding of the emotional toll and burden psoriasis can be on patients
- Non-adherence is a significant barrier to therapeutic success, therefore, discuss acceptable treatment options with the patient
- May take 1 to 2 months to notice improvement
- Multiple trials of medication are usually required
- First-line treatment options for **milder** cases of plaque psoriasis include **topical corticosteroids and vitamin D3 analogues** (as monotherapy or in combination)
- Treatment options for more **severe** plaque psoriasis require systemic therapy, including **immunosuppressives (e.g. methotrexate, cyclosporine), oral retinoids, biologics and PDE-4 inhibitors**
- Educate patients on the risks of teratogenic medications and contraception
- Ensure vaccinations are up-to-date before starting immunosuppressive agents or biologics

CASE 1

RW is a 56-year-old woman presenting to the clinic with a chief complaint of itchy, well-demarcated silvery-pink scales on her arms and trunk. She states she has not recently changed any soaps or detergents but has taken up a new gardening hobby. Her past medication history is significant for hypertension, type 2 diabetes mellitus, and depression. Her current medications include metoprolol 25 mg PO BID, lisinopril 10 mg PO daily, metformin 500 mg PO BID, duloxetine 60 mg PO daily, and bupropion XL 150 mg PO daily. She recently quit smoking after having a history of 1 pack per day for 30 years. She enjoys a glass of wine with dinner. RW states she is quite miserable with the constant itching plaques. Her physician is asking you for a recommendation to help RW.

1. Which of the following is NOT a possible risk factor or trigger for RW presentation?
 - a) History of smoking
 - b) Alcohol consumption
 - c) Taking metoprolol
 - d) History of depression



CASE 1

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2. RW's physician has diagnosed her with mild plaque psoriasis. Which of the following non-pharmacological measures can be recommended to RW to help with her symptoms?
- a) Spending time indoors with air conditioning
 - b) Wearing moisture-wicking clothing while gardening
 - c) Purchasing a dehumidifier for her home
 - d) Avoiding the use of moisturizers on the affected areas



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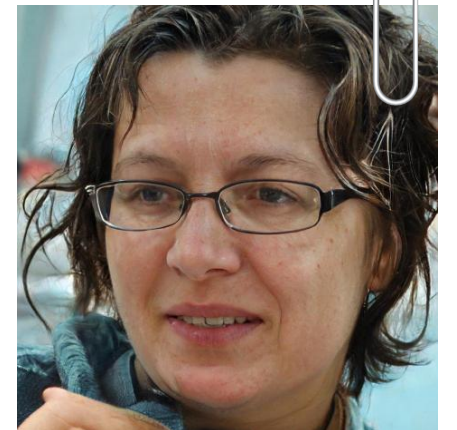
3. Which of the following would be the most appropriate initial therapy for RW's mild plaque psoriasis?
- a) Acitretin and doxycycline
 - b) Secukinumab
 - c) Topical betamethasone
 - d) Methotrexate



CASE 1

RW is a 56-year-old woman presenting to the clinic with a chief complaint of itchy, well-demarcated silvery-pink scales on her arms and trunk. She states she has not recently changed any soaps or detergents but has taken up a new gardening hobby. Her past medication history is significant for hypertension, type 2 diabetes mellitus, and depression. Her current medications include metoprolol 25 mg PO BID, lisinopril 10 mg PO daily, metformin 500 mg PO BID, duloxetine 60 mg PO daily, and bupropion XL 150 mg PO daily. She recently quit smoking after having a history of 1 pack per day for 30 years. She enjoys a glass of wine with dinner. RW states she is quite miserable with the constant itching plaques. Her physician is asking you for a recommendation to help RW.

4. If RW developed a plaque on her face, which of the following topical corticosteroids should be recommended for that area?
- a) Hydrocortisone 1%
 - b) Fluocinonide 0.05%
 - c) Hydrocortisone valerate 0.2%
 - d) Clobetasol propionate 0.05%



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CHANGE LOG

March 2025

- Slide 7: Added scalp psoriasis
- Slide 12: Changed BSA low severity from $\leq 3\%$ to $< 3\%$ in line with cfp guideline and national psoriasis foundation
- Slide 18: For petrolatum, removed wording that says to use with steroids in order to reduce tachyphylaxis as not able to confirm. Added in wording that it is an emollient to moisturize the skin. Also removed wording that lactic acid is 2nd line to salicylic acid as not able to confirm. Added that it may be an option in scalp psoriasis in line with references
- Slide 44: Added references

August 2025

- Formatting, grammar, spelling changes throughout
- Slide 3: Updated legend
- Slides 18, 19: Split non-prescription products into 2 slides, added coal tar
- Slide 20: Added phosphodiesterase-4 inhibitor
- Slide 21: Updated betamethasone and hydrocortisone information based on product availability
- Slide 25: Added roflumilast topical agent
- Slide 35: Added note on trimming nails
- Slide 47: Updated references

October 2025

- Slide 42: Added monitoring parameters slide
- Slide 43: Elaborated on Takeaway slide

February 2026

- Minor formatting changes made
- Slide 40: Added certolizumab use in pregnancy
- Slide 48: Updated references